BODY × ENVIRONMENT DIGITAL TWIN

PranaGuard

A connected respiratory twin.

An honest forecast of what comes next.

COPD exacerbation forecasting · 24 / 48 / 72 hours

24 hours

48 hours

72 hours

[TEAM NAME PENDING]
[COLLEGE / INCUBATOR PENDING]

All patients synthetic.
Not clinically validated.

Respiratory health is a moving picture.

A clinician needs changing patient context, not another disconnected measurement.

The clinical question

Could a personalized combination of history, body signals, symptoms and environment support an earlier review conversation?

The prototype boundary

This build explores the workflow. It does not demonstrate fewer admissions, clinical warning lead time or treatment benefit.

- Synthetic COPD outpatient cohort
- Clinician review, acknowledgement and action logging
- Pune · Mumbai · Delhi · Bengaluru ambient examples
- No clinical participants or hospital partnership claimed

Three horizons. Three actual models.

New synthetic onset in (t, t+h] — not diagnosis, triage or validated lead time.

24

HOUR HORIZON

Independent fitted + calibrated estimate

48

HOUR HORIZON

Independent fitted + calibrated estimate

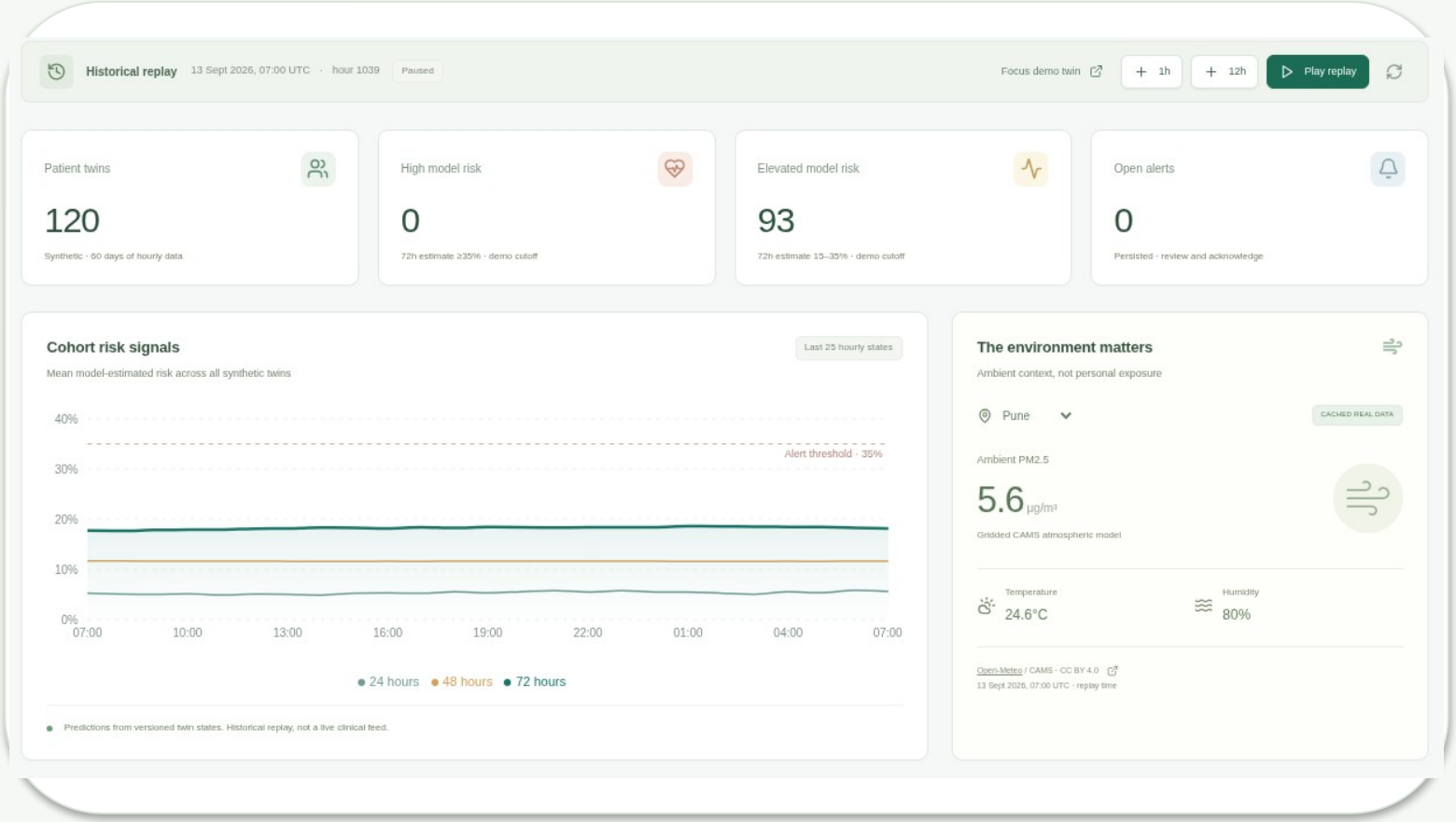
72

HOUR HORIZON

Independent fitted + calibrated estimate

Active episodes, initial72h calibration and incomplete future follow-up are excluded from evaluation. Independent horizons can be non-monotonic.

One clinician workspace, connected end to end.



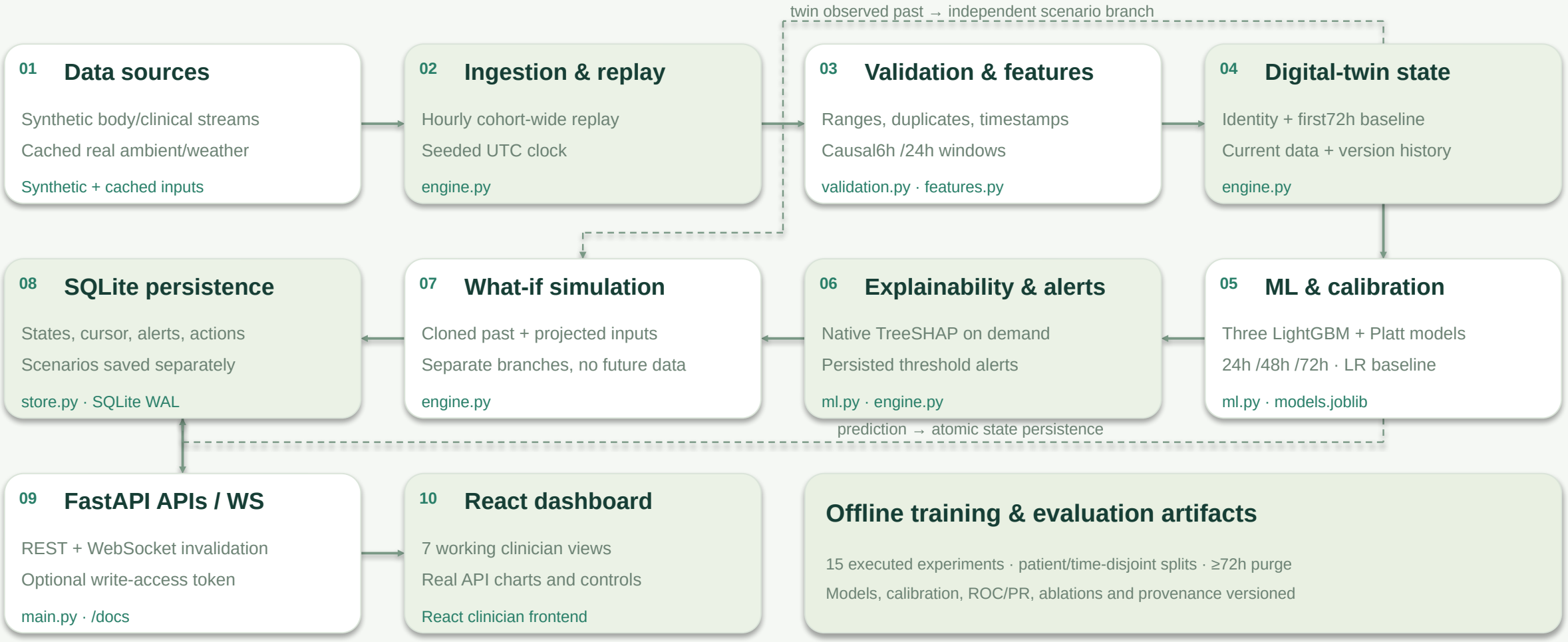
Actual running React frontend screenshot. Values are API-fed synthetic model estimates, not a mockup.

- Cohort risk & source freshness
- Patient-specific baseline and history
- Explanations and isolated scenarios
- Persisted alerts and review trail

A stateful twin — not a static prediction page.

01	**Persistent identity**	Patient-specific state and immutable SQLite versions.
02	**Personal calibration**	First72h baseline precedes fitted and replayed predictions.
03	**Hourly updates**	Observed body/environment context advances deterministically.
04	**Future exploration**	Three fitted risk models plus separate conditional branches.
05	**Auditable review**	Saved explanations-by-state, alerts, actions and scenarios.

A persistent twin, not a static score



Solid arrows: logical capability chain. Dashed: cross-cutting runtime branches. SQLite writes occur on every twin update; explanations and scenarios are API-triggered. All patients synthetic; cached ambient data are historical. The 72h model is weak and its calibration slope is negative.

Know what is real. Know what is synthetic.

SYNTHETIC · clinical/body/outcomes

120 patients × 60 days = 172,800 hourly records

263 stochastic artificial onsets

Seed2026 · no real people or hospital records

Generator is not medically calibrated

REAL CACHED · ambient model products

8 successfully retrieved public API responses

4 cities × air + weather; 1,440 hours/cache

2026-08-01 to 2026-09-29 UTC

Open-Meteo / CAMS · CC BY 4.0

Ambient grids are not personal exposure. Historical products are not live or verified as-issued forecasts. URLs, retrieval times, licenses and SHA-256 checksums are versioned.

The split comes before the score.

No patient identity, future events, future exposure or full-course baseline averages enter the models.

TRAIN

72 patients

14,605 sampled rows

Imputer/scaler fitted here

CALIBRATION

24 patients

1,377 sampled rows

Separate Platt fits here

TEST

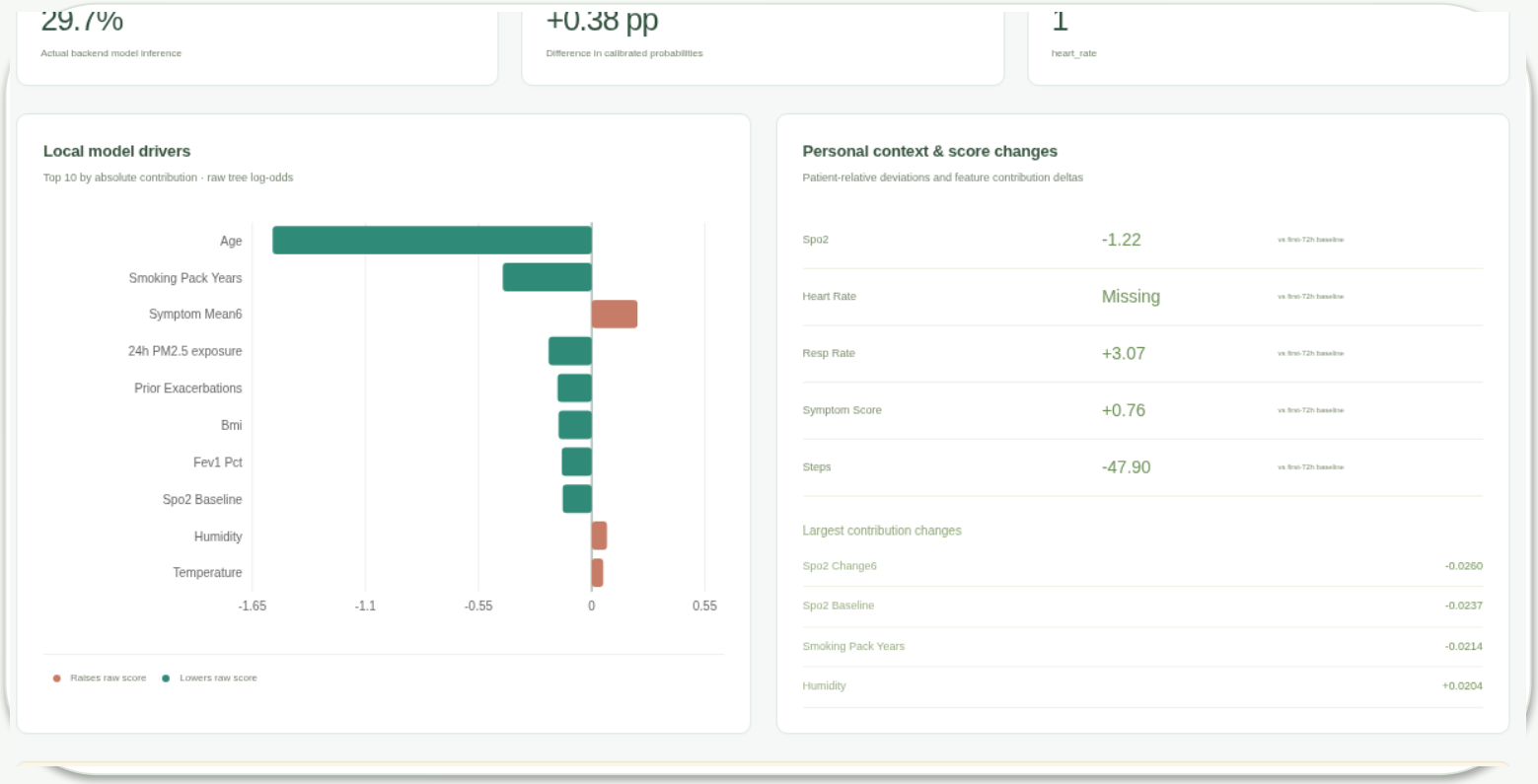
24 patients

4,699 hourly windows

Later dates; untouched for tuning

- Patient-disjoint and calendar-time-disjoint, with ≥72h purges.
- Fixed hyperparameters; LR baseline + 4 LightGBM input groups × 3 horizons.
- Causal trailing features; first72h warm-up; no test-selected threshold or tuning.

Explain the model — and its failure modes.



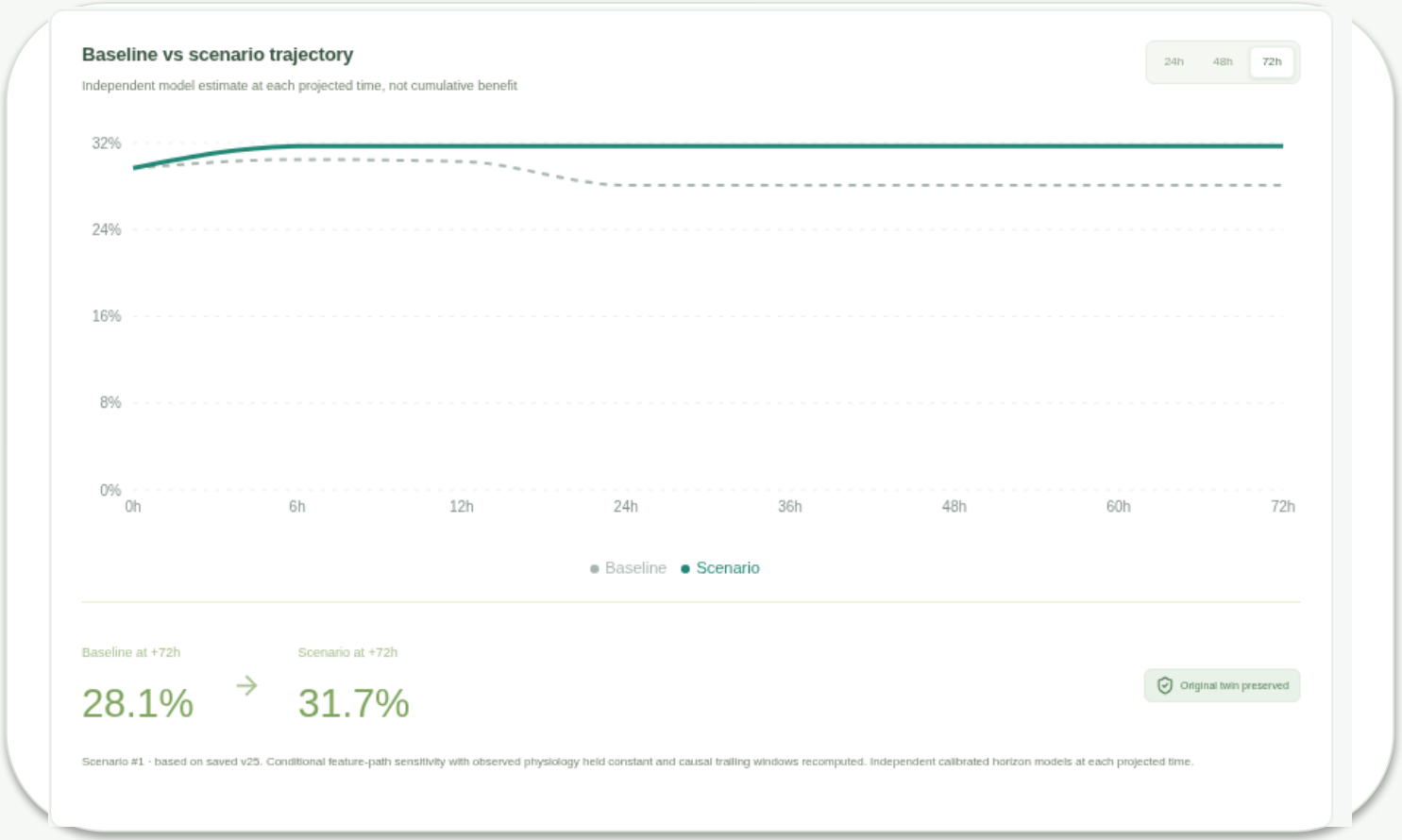
- Native exact TreeSHAP
- Base + contributions = raw log-odds
- Patient-relative deviations and driver deltas

Critical caveat: 72h Platt slope is negative. Raw contribution direction is not calibrated-risk direction. Association ≠ causation.

What changes when assumptions change?

A model sensitivity branch, not a validated treatment or physiological response simulator.

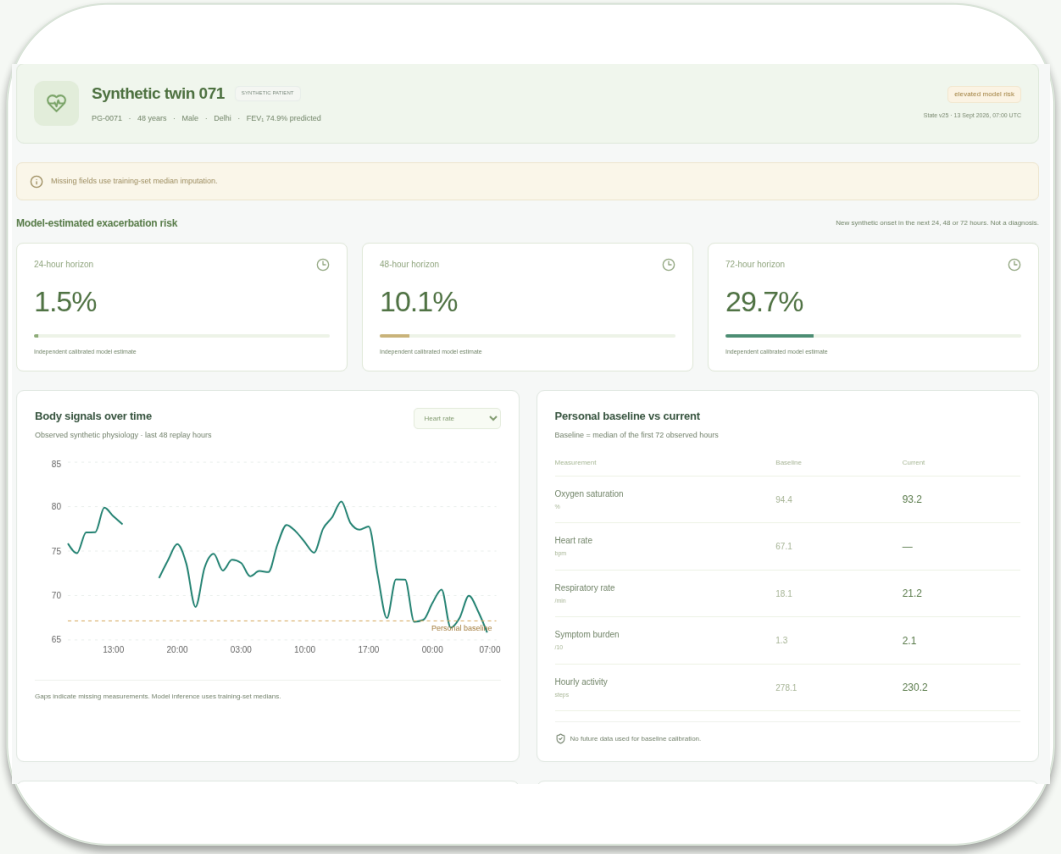
- PM2.5 −80% to +100%
- Adherence feature0–1
- Activity0.5–1.5× steps
- Observed physiology held constant
- Original twin never overwritten



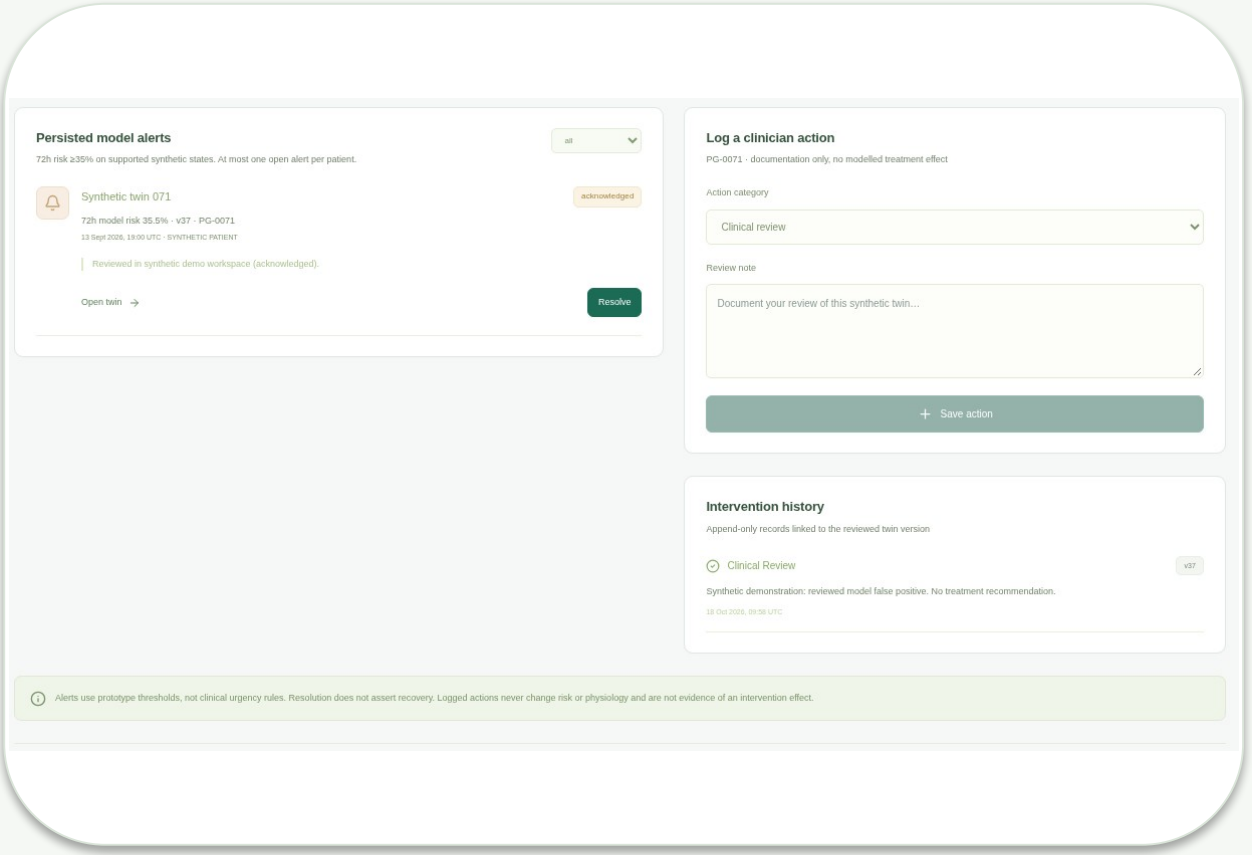
Counterintuitive results are possible. No proven health benefit is implied.

From review to an auditable action.

Open a twin → inspect evidence → review an alert → document an action → reload persisted state.



Actual patient twin, personal baseline and versioned risks



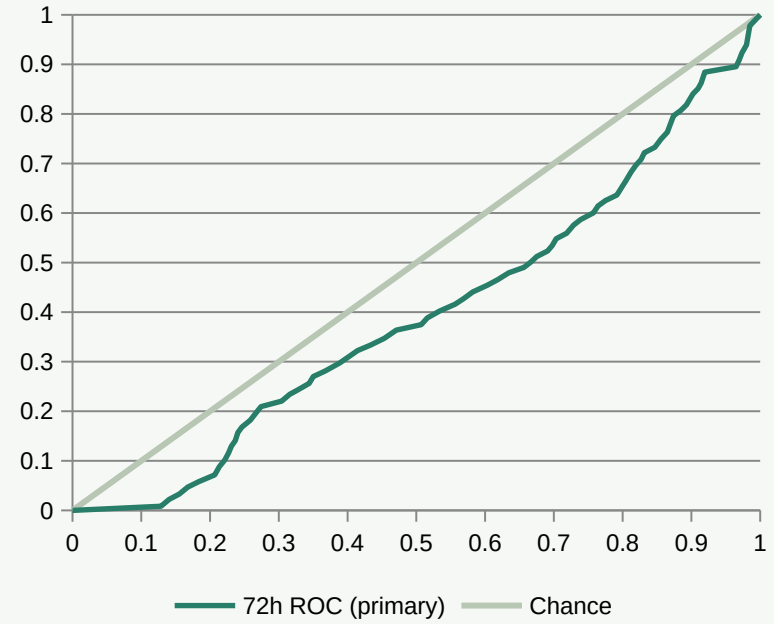
Actual model-generated false-positive alert and persisted review note

The engineering works. The model needs work.

Synthetic held-out results only. Poor72h discrimination and zero sensitivity at the fixed.35 cutoff.

MODEL / HORIZON	AUROC	AUPRC	BRIER
24h · LightGBM	0.6916	0.0882	0.0279
24h · Logistic Reg.	0.6619	0.1993	0.0260
48h · LightGBM	0.6144	0.0968	0.0557
48h · Logistic Reg.	0.6285	0.1841	0.0537
72h · LightGBM	0.3951	0.0593	0.0853
72h · Logistic Reg.	0.6253	0.2281	0.0800

Test:24 patients,4,699 correlated hours.150 patient-cluster bootstrap resamples in JSON.



72h prevalence:0.0773

72h AUPRC:0.0593 (below baseline)

More features did not establish better forecasts.

Same leakage-aware split, fixed model choices and separate calibration for every experiment.

72H INPUT GROUP	AUROC	AUPRC	BRIER
Static Only	0.2197	0.0550	0.0816
Body Stream	0.6358	0.1086	0.0815
Full Environment	0.3119	0.0514	0.0816
Personalized	0.3951	0.0593	0.0853

No improvement claim

Full environment72h is worse than body streams. Personalization does not establish benefit. Calibration instability is retained, not concealed.

Only5 held-out test-window onsets.0 detected at72h ≥35%; median lead time is undefined. All primary horizons have zero sensitivity at that cutoff. Small synthetic samples and shared exposure create major uncertainty.

A reproducible demo, including a false positive.

RESET

1039

24h persisted history · focus PG-0071

ADVANCE

+12h

Replay to1051 · state version37

REVIEW

35.51%

Actual calibrated72h risk crosses.35

DISCLOSE

label = 0

Retrospective score-selected false positive

Show the full workflow

Play/pause → reset → +12h → explanation → acknowledge/log/resolve → scenario → reload → actual validation.

Clinical claims require clinical evidence.

- Synthetic generator and labels are not clinically calibrated.
- 72h model/calibration are unstable; no useful warning is established.
- Ambient grids ≠ personal dose; scenarios are non-causal.
- Local demo token ≠ production identity, privacy or governance.

Next, before real-world use

Ethics/consent + adjudicated real outcomes

New pre-registered development data

External prospective patient/time/site validation

Calibration, fairness and uncertainty assessment

Secure ingestion and clinician safety governance

Build the workflow. Measure honestly.

Persistent state. Real inference. Transparent limitations.

120 synthetic twins · 60 days · 3 horizons

Reproducible experiments and a working clinician workspace.

Team / institution details: PENDING

Public GitHub + signed-out file checks: PENDING

Actual20min Unlisted YouTube upload: PENDING

NOT FULLY SUBMISSION-READY until the mandatory external items are genuinely complete.